

Development of a Secure Web-Based Medical Image Management System Using Steganography

Protecting patient diagnosis data inside medical images without visible distortion

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Presentation Outline

Following the format supplied for the real/internal defense



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Introduction

Why secure medical image handling matters

Context

Medical images often move with supporting diagnosis notes, patient identifiers, and clinical comments. If shared carelessly, these details can expose confidential health information.

Steganography

Steganography hides sensitive text inside an image file. Unlike visible encryption warnings, the image remains visually normal while carrying the embedded message.

Project Focus

This project implements a secure web-based system that embeds and retrieves patient diagnosis text from medical images while preserving image quality.

Review of Literature and Related Works

Client-supplied literature review format: Objective • Methodology • Result • Contribution

Author	Objective	Methodology	Result	Contribution
Cheddad et al.	Survey image steganography	Comparative review	Capacity, security, imperceptibility trade-offs	Guides algorithm selection
Morkel et al.	Explain steganography principles	Conceptual analysis	Hidden communication protects message presence	Supports use in sensitive data transfer
Sharma & Prabha (2025)	Fast clinical image embedding	LSB/client-side image processing	Efficient execution for real-time use	Supports browser-based workflow
Kouadri et al. (2025)	Medical image quality preservation	PSNR/MSE evaluation	High PSNR retains diagnostic value	Basis for quality validation

Gap addressed: a practical web system combining LSB steganography, authentication, and audit logging for medical image workflows.

Motivation

The problem that led to the project



Data privacy risk

Diagnosis notes can be exposed when images are transferred through general channels or unsecured storage.

Image integrity

Medical images must remain diagnostically useful; security must not visibly damage regions of interest.

Accountability

Clinical systems need traceability, so encryption and decryption events should be logged with user and file context.

Objectives

What the system was designed to achieve

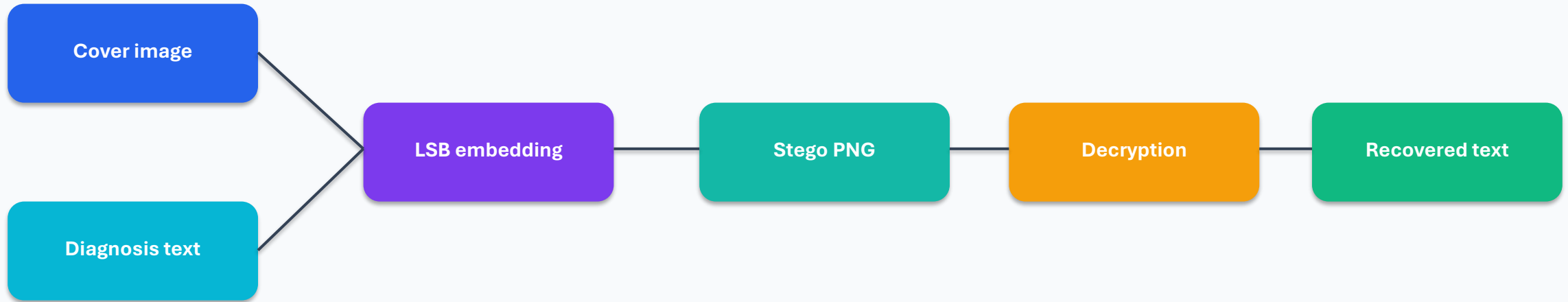
- Develop a secure web-based dashboard for authenticated users.
- Embed patient diagnosis text into medical images using LSB steganography.
- Recover hidden diagnosis text accurately during decryption.
- Preserve visual quality so the stego-image remains medically usable.
- Record encryption/decryption activities for audit and accountability.

Success Criteria

- Hidden text recovered without corruption
- Minimal visual difference between cover and stego-image
- Responsive browser processing
- Event logs stored through Prisma/SQlite

Methodology

LSB embedding and recovery workflow



Embedding

Text is converted to binary and placed in the least significant bits of RGB pixel channels.

Recovery

Bits are extracted until the terminator is found, reconstructing the diagnosis text.

Audit

Each operation is logged with file context, user, status, and timestamp.

System Architecture and Core Modules

Implementation stack and responsibilities

Authentication Dashboard

Protected routes for dashboard, encryption lab, decryption lab, and settings using Better Auth.

Steganography Engine

HTML5 Canvas extracts pixel data; LSB algorithm embeds and retrieves diagnosis text from image pixels.

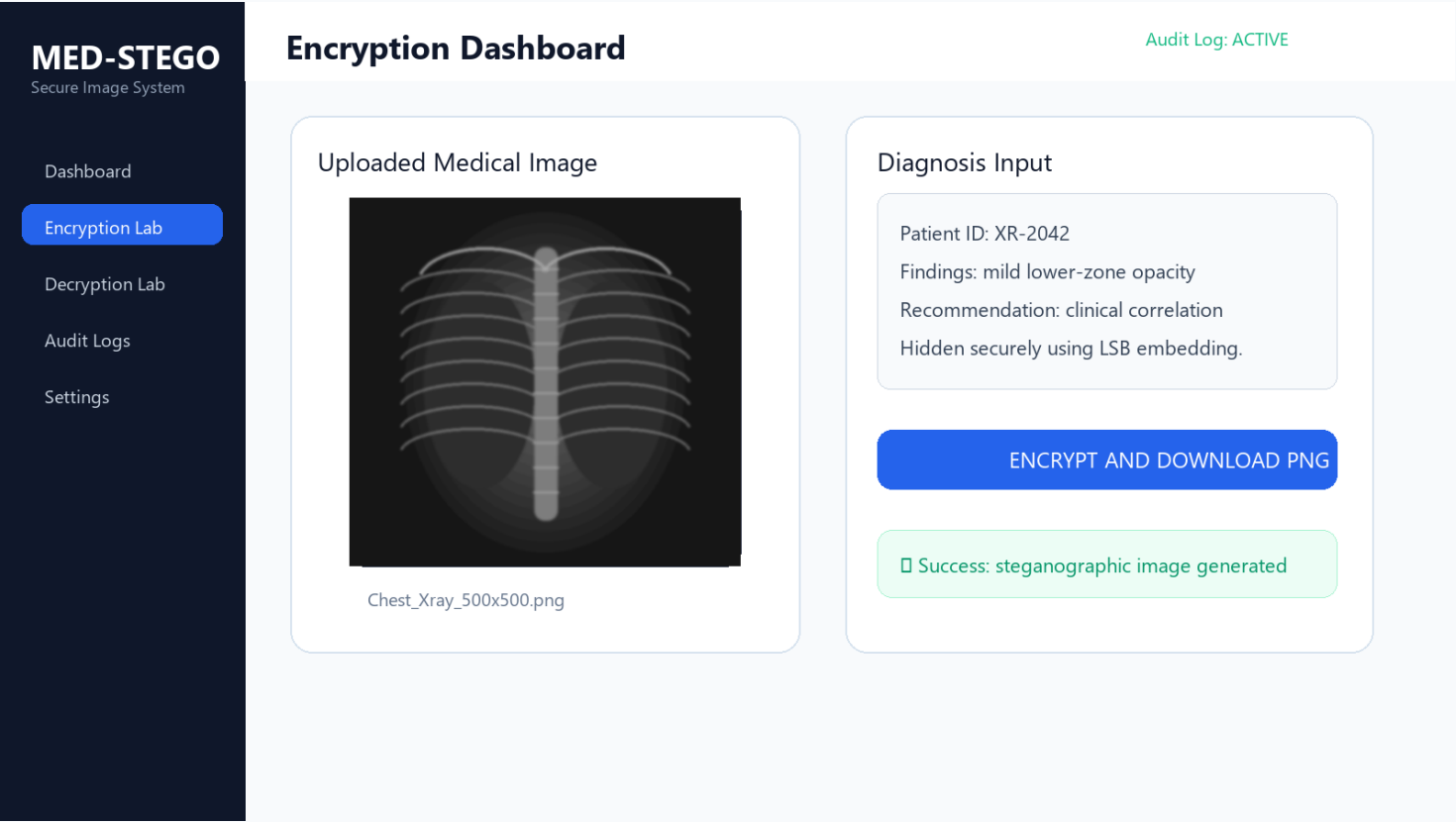
Audit Logging

Prisma ORM stores encryption and decryption events in SQLite for accountability and later review.



Result and Discussion

Encryption dashboard output



Test Case 1: Encryption

A standard chest X-ray image and diagnosis text were successfully processed into a downloadable stego PNG.

Visual Quality

The output image remained visually indistinguishable from the original cover image.

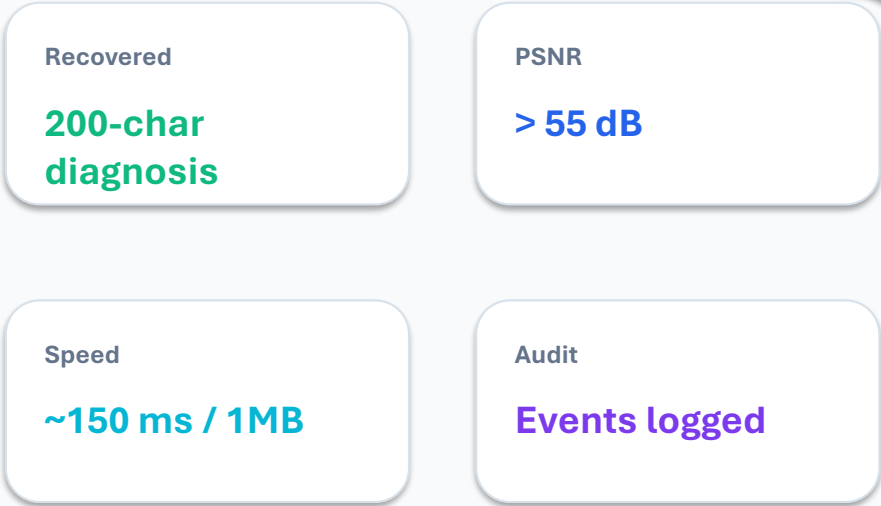
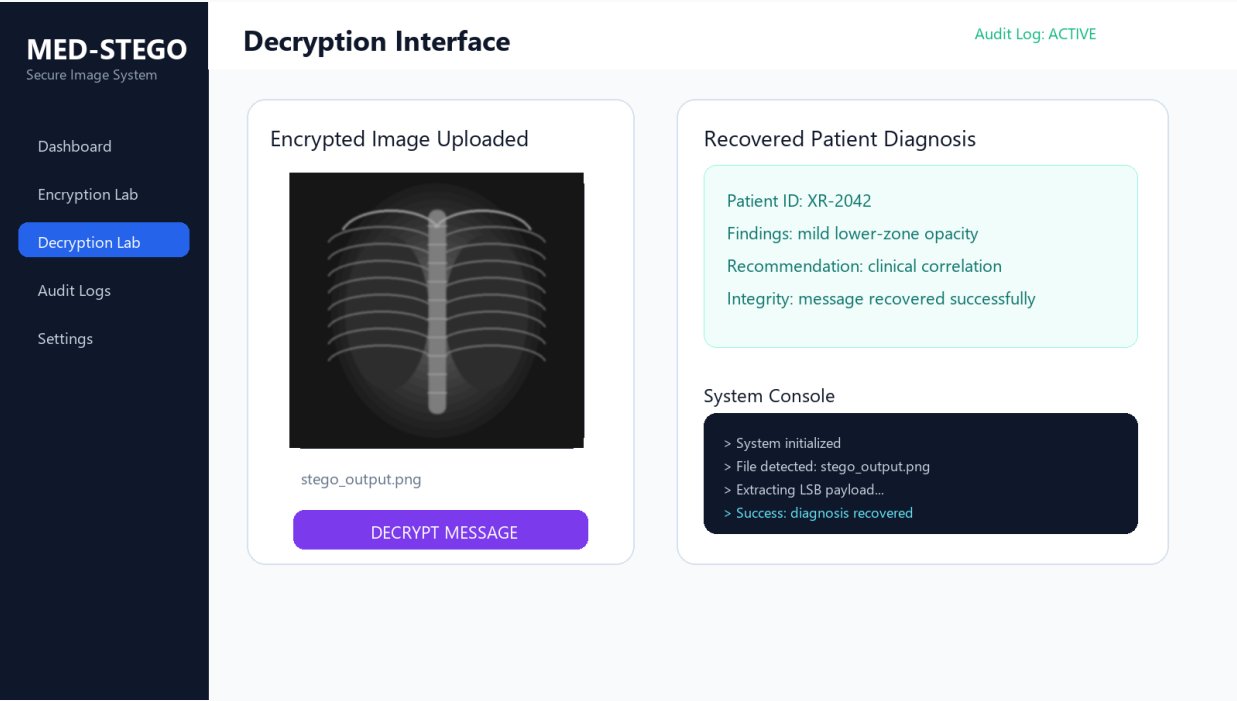
User Feedback

Real-time status indicators confirmed processing and success states.

Figure 4.1: Screenshot of the Encryption Dashboard showing the uploaded X-ray and the diagnosis input field

Functional Testing and Performance

Decryption, audit logging, and quality metrics



Discussion

The results indicate that the system balances security, usability, and image imperceptibility for medical image workflows.

Figure 4.2: Screenshot of the Decryption Interface displaying the recovered patient diagnosis

Contribution to Knowledge

What this project adds

Medical-data hiding workflow

Demonstrates a usable path for pairing diagnosis text with medical image files without obvious external metadata.

Browser-side processing

Uses client-side Canvas processing, reducing the need to upload raw patient notes to a server before embedding.

Security + accountability

Combines steganography with authenticated access and audit logging in one practical web prototype.

Quality-aware evaluation

Uses imperceptibility, PSNR, speed, and functional recovery as defense-ready validation points.

Conclusion and Recommendation

Summary and next steps

Conclusion

The developed system successfully embeds and retrieves diagnosis data inside medical images using LSB steganography. Testing showed accurate recovery, high visual imperceptibility, responsive processing, and traceable audit logs.

Recommendations

- Extend testing with larger DICOM datasets
- Combine LSB with stronger cryptographic encryption
- Add role-based clinical access controls
- Deploy with PostgreSQL and secure cloud storage
- Run usability testing with medical users

References

Selected literature and project sources

- Cheddad, A., Condell, J., Curran, K., & Mc Kevitt, P. (2010). Digital image steganography: Survey and analysis of current methods.
- Morkel, T., Eloff, J. H. P., & Olivier, M. S. (2005). An overview of image steganography.
- Provos, N., & Honeyman, P. (2003). Hide and seek: An introduction to steganography.
- Sharma, R., & Prabha, S. (2025). Client-side image steganography for efficient clinical data protection.
- Kouadri, A., et al. (2025). Quality evaluation of steganographic medical images using MSE and PSNR metrics.
- Project source files: Chapter Four.docx, README.md, and implemented Next.js application modules.

Thank you